

where ℓ is some convex loss function and the q_i 's are data pruning parameters ($q_i \in \{0, 1\}$), indicating whether to select or drop the i^{th} synthetic sample $(\tilde{\mathbf{x}}_i, \tilde{y}_i)$. In particular, the q_i 's are Bernoulli random variables conditionally on $\tilde{y}_i \neq \bar{y}_i$ or $\tilde{y}_i = \bar{y}_i$ (we recall that $\bar{y}_i$'s denote the true labels of the synthetic samples) with conditional probabilities

$$\rho := \mathbb{P}\{q_i = 1 \mid \tilde{y}_i \neq \bar{y}_i\}, \quad \phi := \mathbb{P}\{q_i = 1 \mid \tilde{y}_i = \bar{y}_i\}, \quad (5)$$

which control the pruner accuracy (as discussed in (Feng et al., 2024)). As we mentioned previously, we suppose training on $n \geq \hat{n}$ real data, modeling a situation where new real samples are available with $\hat{n}$ controlling the generative model quality in generating faithful synthetic features¹.

L^2 -loss. In the remainder of the paper we take ℓ to be the regularized squared loss as it allows us to obtain a closed-form solution for the optimization problem in equation 4, hence, a more tractable analysis. Specifically, we take $\ell(\mathbf{x}, y; \mathbf{w}) = (\mathbf{w}^\top \mathbf{x} - y)^2 + \gamma \|\mathbf{w}\|^2$ where $\gamma \geq 0$ is a regularisation parameter, which yields the following closed-form solution

$$\mathbf{w} = \frac{1}{N} \mathbf{Q} \mathbf{X} \mathbf{y}, \quad \mathbf{Q} = \left(\frac{1}{N} \mathbf{X} \mathbf{X}^\top + \gamma \mathbf{I}_p \right)^{-1}. \quad (6)$$

where $N = n + m$, the matrix $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_n, q_1 \tilde{\mathbf{x}}_1, \dots, q_m \tilde{\mathbf{x}}_m) \in \mathbb{R}^{p \times N}$ is the concatenation of both real and (pruned) synthetic features, and the vector $\mathbf{y} = (y_1, \dots, y_n, \tilde{y}_1, \dots, \tilde{y}_m) \in \mathbb{R}^N$ is the concatenation of real and (noisy) synthetic labels.

4 MAIN RESULTS

In this section, we present and discuss the main results obtained through the analysis of the classifier model defined in equation 6. We start by specifying the supposed growth rate assumptions.

Assumption 4.1 (Growth Rate). *We consider a high-dimensional regime where $p, n, \hat{n}, m \rightarrow \infty$ and we recall $N = n + m$ such that:*

$$1) \frac{p}{N} \rightarrow \eta \in [0, \infty), \quad 2) \frac{p}{\hat{n}} \rightarrow \hat{\eta} \in [0, \infty), \quad 3) \frac{n}{N} \rightarrow \pi \in [0, 1], \quad 4) \|\boldsymbol{\mu}\| = \mathcal{O}(1).$$

Role of the assumptions. The above assumptions are central to understanding the nuances between real and synthetic data (as constructed above) in a high-dimensional regime. Essentially,

- Assumptions 1), 2), and 3) define the scaling of data dimension p and the different sample sizes (n real data, m synthetic data, and $\hat{n}$ real samples used to train the generative model). In particular, we suppose that all these dimensions scale linearly relative to each other, which corresponds to the classical RMT regime. This setting is more general than the infinite sample size regime in the sense that the former can be recovered by taking $\eta, \hat{\eta} \rightarrow 0$. Specifically, the parameter $\hat{\eta}$ controls the generative model quality, where lower values indicate better generative model quality. Plus, the parameter π corresponds to the proportion of the real samples in the data mixture. For instance, $\pi = 0$ models a setting where the training is done only on synthetic samples, and $0 < \pi < 1$ highlights the fact that the number n of real and m of synthetic samples are of the same order, therefore, making our results scalable to any possible proportion π .
- The fourth condition about the magnitude of the mean vector $\boldsymbol{\mu}$ reflects the fact that the classification problem should neither be trivial ($\|\boldsymbol{\mu}\| \gg 1$) nor impossible ($\|\boldsymbol{\mu}\| \rightarrow 0$) as the dimension of data grows large. For instance, assuming $\|\boldsymbol{\mu}\|$ of order $\mathcal{O}(\sqrt{p})$ would not be relevant as $p \rightarrow \infty$ since the classification problem becomes trivial in this regime. We refer the reader to (Couillet & Benaych-Georges, 2016) for a more general formulation and justifications of this assumption under an extended k -class Gaussian mixture model.

Having stated the main assumptions, we are now in place to present our main technical findings on the performance of the classifier model trained on a mixture of real and synthetic data. As a corollary, we also cover the case where the model is trained solely on synthetic data and showcase a generalization of the result obtained by Feng et al. (2024).

¹Technically, our results hold irrespective of the statistical dependencies between the data used to train the generative model in equation 2 or the classifier in equation 6.

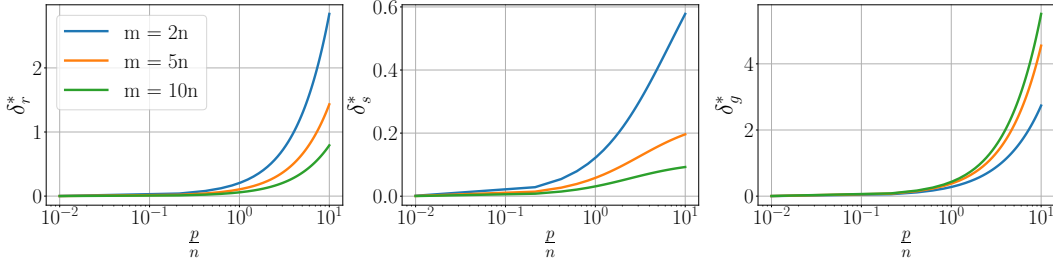


Figure 2: Behavior of $(\delta_r^*, \delta_s^*, \delta_g^*)$ in terms of the ratio $\frac{p}{n}$. For small ratio $\frac{p}{n}$, the values of $\delta_r^*, \delta_s^*, \delta_g^*$ are close to 0. $(\delta_r^*, \delta_s^*, \delta_g^*)$ are computed by iterating the system 7 starting from random values.

4.1 PARTIALLY SYNTHETIC: MIXTURE OF REAL AND SYNTHETIC DATA

We start by analyzing the general case of training on a mix of real and synthetic data. As we described in the previous section, the statistics of synthetic data are empirical estimates of the ones of real data. Under Assumption 4.1, the estimation of μ with $\hat{\mu}$ remains consistent, while the estimation of the underlying real data covariance (i.e., $\mathbf{I}_p$ in our setting) with $\hat{\mathbf{C}}$ is inconsistent as we previously discussed. As a result, studying the theoretical performance of the classifier in equation 6 demands deploying tools from random matrix theory that refines the estimation of scalar quantities depending on large random matrices. In our case, the scalar quantity of interest corresponds to the model's accuracy which depends on the random matrices $\hat{\mathbf{C}}$ and $\mathbf{X}\mathbf{X}^\top$ as per equation 6.

In our analysis of the classifier's theoretical performance, we found that the effect of high-dimension (and that of distribution shift between real and synthetic samples) is described by three scalar quantities $(\delta_r^*, \delta_s^*, \delta_g^*)$ which are defined as the unique solution of the following fixed point system which is derived from Lemma D.1 in the Appendix.

$$\begin{cases} \delta_g = \frac{\alpha(1-\pi)}{1+\delta_s} \cdot \frac{\hat{\eta}}{\gamma + \frac{\pi}{1+\delta_r} + \frac{\alpha(1-\pi)}{(1+\delta_s)(1+\delta_g)}}, \\ \delta_r = \frac{\eta}{\hat{\eta}} \cdot \frac{1+\delta_s}{\alpha(1-\pi)} \delta_g, \\ \delta_s = \frac{\alpha\delta_r}{1+\delta_g}. \end{cases} \quad (7)$$

where $\alpha = \phi(1-\varepsilon) + \rho\varepsilon$. These quantities will be used subsequently in our results. Intuitively, δ_r^* captures the contribution of real data, δ_s^* corresponds to the contribution of synthetic data, and δ_g^* corresponds to the influence of the generative model. In an infinite sample size regime where $n, m, \hat{n} \rightarrow \infty$ while the dimension p is kept fixed, $(\delta_r^*, \delta_s^*, \delta_g^*) = (0, 0, 0)$ as per Fig. 2, while under Assumption 4.1 these quantities are non zero yielding a counterintuitive behavior in high-dimension. For convenience, we further define a set of scalar quantities that will prove usefull in the next result.

$$\begin{aligned} \alpha &= \mathbb{E}[q_i] = \phi(1-\varepsilon) + \rho\varepsilon, \quad \lambda = \mathbb{E}[q_i \tilde{y}_i] = \phi(1-\varepsilon) - \rho\varepsilon, \\ a &= \frac{\pi}{1+\delta_r^*} + \frac{\alpha(1-\pi)}{1+\delta_s^*}, \quad b = \gamma + \frac{\pi}{1+\delta_r^*} + \frac{\alpha(1-\pi)}{(1+\delta_s^*)(1+\delta_g^*)}, \quad c = \frac{\pi}{1+\delta_r^*} + \frac{\lambda(1-\pi)}{1+\delta_s^*}, \\ a_1 &= \frac{\pi\eta}{(1+\delta_r^*)^2 h_2 b^2}, \quad b_1 = \frac{\alpha(1-\pi)\eta}{(1+\delta_s^*)^2 (1+\delta_g^*)^2 h_2 b^2}, \quad b_2 = \frac{\alpha(1-\pi)\eta}{(1+\delta_s^*)^2 (1+\delta_g^*)^4 h_2 b^2}, \\ h_1 &= 1 - a_1 - b_2, \quad h_2 = 1 - \left(\frac{\alpha(1-\pi)}{(1+\delta_s^*)(1+\delta_g^*)} \right)^2 \frac{\hat{\eta}}{b^2}. \end{aligned}$$

The first set of parameters $(\alpha, \lambda, a, b, c)$ pop out from the expectation of the classifier's decision function while the remaining quantities are related to second-order statistics. Essentially, the main relevant quantities to our analysis are $\hat{\eta}$ and ε which characterize the quality of synthetic data, with ϕ and ρ characterizing the verification process. In an idealized scenario, we would have $\hat{\eta} = \varepsilon = 0$ which reflects the fact that there is no distribution shift nor label noise, while $\phi = 1 - \rho = 1$ corresponds to a perfect (oracle) verification process. Our main goal is to study how these parameters influence the classifier's performance hence providing the conditions that make synthetic data relevant for performance boost. The main result brought by this paper is therefore stated as follows.

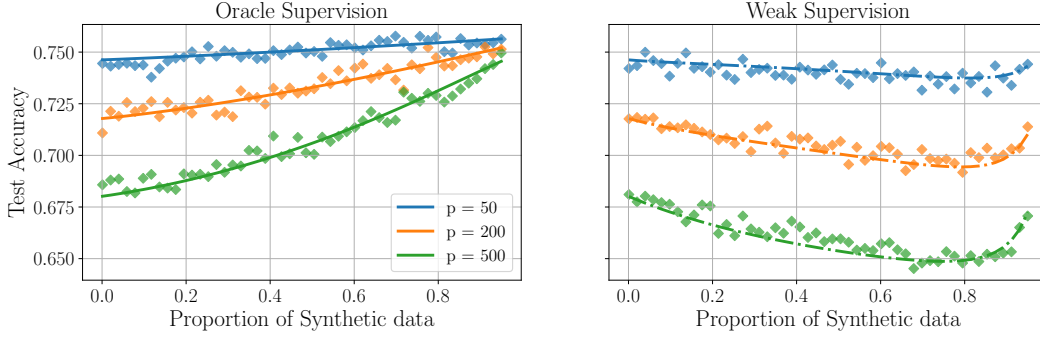


Figure 3: Scatter plots correspond to empirical test accuracy while lines correspond to the theoretical counterpart as per Theorem 4.2. The parameters used in this experiments are: $n = \hat{n} = 1000$, $\|\mu\| = 0.7$ and $\gamma = 1$, $(\rho, \phi) = (0, 1)$ for Oracle supervision and $(\rho, \phi) = (1, 0.5)$ for the Weak supervision. The parameter ε is variable depending on the proportion of synthetic data by taking it equal to the misclassification error corresponding to training a classifier on synthetic data only. As theoretically anticipated, a boost of performance is observed with synthetic data supervision while distribution shift affects negatively the performance.

Theorem 4.2 (Theoretical performance). *Let w be the Ridge classifier as defined in equation 6 and suppose that Assumption 4.1 holds. The decision function $w^\top x$, on some (real) test sample $x \in \mathcal{C}_a$, with corresponding label $y = (-1)^a$ and independent of $\mathbf{X}$, satisfies*

$$w^\top x \xrightarrow{\mathcal{D}} \mathcal{N}(y \cdot \mu, \nu - m^2),$$

where $\mu = \frac{c\|\mu\|^2}{b+a\|\mu\|^2}$ and

$$\nu = \frac{c\|\mu\|^2}{h_1(b+a\|\mu\|^2)^2} \left(c(1+b_1-b_2)\|\mu\|^2 + \frac{c}{h_2} - 2 \left(a_1 + \frac{\lambda b_1}{\alpha} \right) (b+a\|\mu\|^2) \right) + \frac{a_1+b_1}{h_1}.$$

Moreover, the asymptotic test accuracy of the classifier is given by $\Phi((\nu - m^2)^{-\frac{1}{2}}m)$ where

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{t^2}{2}} dt.$$

Theorem 4.2 states that the decision function of the classifier in equation 6 is asymptotically equivalent to the thresholding of two monivariate Gaussian random variables with respective means μ and $-\mu$ and standard deviation ν , where the statistics μ and ν are expressed in terms of the scalar quantities defined above. Here, μ represents the signal strength while ν highlights the classifier's uncertainty or dispersion. To provide some insights into the implications of this theorem, we start by examining it in a low-dimensional regime where p is kept fixed while $n, m, \hat{n} \rightarrow \infty$. In this case, we have $\eta, \hat{\eta} \rightarrow 0$ and $\delta_r^*, \delta_s^*, \delta_g^* \rightarrow 0$ which yields

$$a = \pi + \alpha(1 - \pi), \quad b = \gamma = \pi + \alpha(1 - \pi), \quad c = \pi + \lambda(1 - \pi),$$

and $a_1 = b_1 = b_2 = 0$ with $h_1 = h_2 = 1$. As such, the accuracy of the classifier increases with λ , i.e., when the synthetic labels are verified (large $\frac{\phi}{\rho}$) or less noisy (small ε). This is in line with the findings of Feng et al. (2024) while extended by our result to training on a mix of real and synthetic data. However, when the dimension scales linearly with the different sample sizes, the values of $\delta_r^*, \delta_s^*, \delta_g^* \not\rightarrow 0$ yielding a lower signal strength μ and higher variance ν^2 . This highlights the fact that in high-dimension, even if the synthetic labels are not noisy or equivalently well verified, there is a performance drop due to the feature distribution shift between real and synthetic data.

Fig. 3 depicts the empirical test accuracy and the theoretical prediction as per Theorem 4.2 when varying the proportion of synthetic data. As theoretically anticipated, adding synthetic data does not boost the classifier's performance unless it is verified accurately (oracle supervision versus weak supervision). Moreover, our results show the effect of the distribution shift which heavily affects performance in the case of weak supervision (Fig. 3 right).